Local Nonlinear Projection-Based Reduced-Order Models

Sebastian Ares de Parga[3], Radek Tezaur[1], Charbel Farhat[1,2]

[1]Department of Aeronautics and Astronautics
[2]Institute for Computational and Mathematical Engineering
Stanford University, Stanford, CA 94305, USA
[3]CIMNE, Barcelona 08034, Spain

April 6, 2026

Global ROM Storyline: LSPG + Gauss-Newton

Given the implicit HDM residual $\mathbf{r}(\mathbf{u}) = \mathbf{0}$, define a manifold

$$\mathbf{u} \approx \mathbf{u}_{\mathrm{ref}} + \widetilde{\mathbf{u}}(\mathbf{q}).$$

LSPG solves at each time step:

$$\mathbf{q}^{\star} = \arg\min_{\mathbf{q}} \ \frac{1}{2} \left\| \mathbf{r}(\mathbf{u}_{\mathrm{ref}} + \widetilde{\mathbf{u}}(\mathbf{q})) \right\|_2^2 .$$

With Gauss-Newton (current iterate $\mathbf{q}^{(k)}$), the increment $\Delta\mathbf{q}$ is obtained from

$$\left(\boldsymbol{\Psi}^T \mathbf{J}^T \mathbf{J} \boldsymbol{\Psi}\right) \Delta\mathbf{q} = -\boldsymbol{\Psi}^T \mathbf{J}^T \mathbf{r},$$

where

$$\boldsymbol{\Psi} = \frac{\partial \widetilde{\mathbf{u}}}{\partial \mathbf{q}}, \qquad \mathbf{J} = \frac{\partial \mathbf{r}}{\partial \mathbf{u}}, \qquad \mathbf{q}^{(k+1)} = \mathbf{q}^{(k)} + \Delta\mathbf{q}.$$

Global Linear PROM

State ansatz:

$$\mathbf{u} \approx \mathbf{u}_{\mathrm{ref}} + \mathbf{V}\mathbf{q}, \qquad \mathbf{V} \in \mathbb{R}^{N \times n},\ n = 151.$$

Hence

$$\widetilde{\mathbf{u}}(\mathbf{q}) = \mathbf{V}\mathbf{q}, \qquad \boldsymbol{\Psi} = \frac{\partial \widetilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V}.$$

Gauss-Newton system:

$$\left(\mathbf{V}^T \mathbf{J}^T \mathbf{J} \mathbf{V}\right) \Delta \mathbf{q} = -\mathbf{V}^T \mathbf{J}^T \mathbf{r}.$$

Global QPROM

Quadratic-manifold ansatz:

$$\mathbf{u} \approx \mathbf{u}_{\mathrm{ref}} + \mathbf{V}\mathbf{q} + \mathbf{H}\left(\mathbf{q} \otimes \mathbf{q}\right), \qquad n = 19.$$

Thus

$$\widetilde{\mathbf{u}}(\mathbf{q}) = \mathbf{V}\mathbf{q} + \mathbf{H}\left(\mathbf{q} \otimes \mathbf{q}\right),$$

and the tangent depends on the current reduced state:

$$\boldsymbol{\Psi}(\mathbf{q}) = \frac{\partial \widetilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V} + \frac{\partial}{\partial \mathbf{q}}\left[\mathbf{H}\left(\mathbf{q} \otimes \mathbf{q}\right)\right].$$

Global PROM-GPR

Split basis and reduced coordinates:

$$\mathbf{V}_{\mathrm{tot}} = [\mathbf{V}, \bar{\mathbf{V}}], \qquad \mathbf{q}_{\mathrm{tot}} = \begin{bmatrix} \mathbf{q} \\ \bar{\mathbf{q}} \end{bmatrix}.$$

$$\bar{\mathbf{q}} = \mathcal{N}_{\mathrm{GPR}}(\mathbf{q}), \qquad n = 10,\ \bar{n} = 141.$$

State ansatz:

$$\mathbf{u} \approx \mathbf{u}_{\mathrm{ref}} + \widetilde{\mathbf{u}}(\mathbf{q}),$$

$$\widetilde{\mathbf{u}}(\mathbf{q}) = \mathbf{V}_{\mathrm{tot}}\mathbf{q}_{\mathrm{tot}} = \mathbf{V}\mathbf{q} + \bar{\mathbf{V}}\mathcal{N}_{\mathrm{GPR}}(\mathbf{q}).$$

Effective tangent for Gauss-Newton:

$$\boldsymbol{\Psi}(\mathbf{q}) = \frac{\partial \widetilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V} + \bar{\mathbf{V}}\frac{\partial \mathcal{N}_{\mathrm{GPR}}}{\partial \mathbf{q}}.$$

Global PROM-POD-DL

Latent reduced coordinate $\mathbf{z} \in \mathbb{R}^5$ with learned map:

$$\mathbf{q}_{\mathrm{tot}} = \mathcal{N}_{\mathrm{DL}}(\mathbf{z}), \qquad \mathbf{q}_{\mathrm{tot}} \in \mathbb{R}^{151}.$$

State reconstruction:

$$\mathbf{u} \approx \mathbf{u}_{\mathrm{ref}} + \widetilde{\mathbf{u}}(\mathbf{z}),$$

$$\widetilde{\mathbf{u}}(\mathbf{z}) = \mathbf{V}_{\mathrm{tot}}\mathbf{q}_{\mathrm{tot}} = \mathbf{V}_{\mathrm{tot}}\mathcal{N}_{\mathrm{DL}}(\mathbf{z}), \qquad \mathbf{V}_{\mathrm{tot}} \in \mathbb{R}^{N\times 151}.$$

Effective tangent for Gauss-Newton:

$$\boldsymbol{\Psi}(\mathbf{z}) = \frac{\partial \widetilde{\mathbf{u}}}{\partial \mathbf{z}} = \mathbf{V}_{\mathrm{tot}} \frac{\partial \mathcal{N}_{\mathrm{DL}}}{\partial \mathbf{z}}.$$

Local ROMs: piecewise manifolds

Partition the training snapshots into clusters $k = 1, \ldots, N_c$ and build one local trial manifold per cluster:

$$\mathbf{u} \approx \Phi_k(\mathbf{q}_k).$$

Examples used in this work:

$$\Phi_k^{\mathrm{lin}}(\mathbf{q}) = \mathbf{u}_{0,k} + \mathbf{V}_k \mathbf{q},$$

$$\Phi_k^{\mathrm{quad}}(\mathbf{q}) = \mathbf{u}_{0,k} + \mathbf{V}_k \mathbf{q} + \mathbf{H}_k(\mathbf{q} \otimes \mathbf{q}),$$

$$\Phi_k^{\mathrm{gpr}}(\mathbf{q}) = \mathbf{u}_{0,k} + \mathbf{V}_k \mathbf{q} + \bar{\mathbf{V}}_k \mathcal{N}_{\mathrm{GPR},k}(\mathbf{q}).$$

Online workflow

1. Select the active cluster.
2. If the cluster changes, transfer the current state to the new local manifold.
3. Solve the local LSPG problem in the active manifold.

In the next slides, we focus only on steps 1 and 2.

Cheap Cluster Selection (1/2)

Current active cluster k^-, current local coordinates $\mathbf{q}_{k^-}$:

$$\mathbf{u}^s = \mathbf{u}_{0,k^-} + \mathbf{V}_{k^-}\mathbf{q}_{k^-}.$$

Choose the next cluster by nearest centroid:

$$k^+ = \arg \min_{\ell \in \{1,\ldots,N_c\}} \|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2.$$

Distance expansion:

$$\|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2 = \|\mathbf{q}_{k^-}\|_2^2 + 2(\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell})^T \mathbf{V}_{k^-}\mathbf{q}_{k^-} + \|\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}\|_2^2.$$

Offline precompute:

$$\mathbf{g}_{k^-,\ell} := \mathbf{V}_{k^-}^T(\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}), \qquad d_{k^-,\ell} := \|\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}\|_2^2.$$

Cheap Cluster Selection (2/2)

Online selector:

$$k^+ = \arg\min_{\ell} \left(2\, \mathbf{g}_{k^-,\ell}^T \mathbf{q}_{k^-} + d_{k^-,\ell} \right),$$

since $\|\mathbf{q}_{k^-}\|_2^2$ is independent of ℓ.

Switch decision

$$k^+ = k^- \;\Rightarrow\; \text{no transfer}, \qquad k^+ \neq k^- \;\Rightarrow\; \text{solve switch projection}.$$

Current strategy

- Local PROM: selector used directly.
- Local QPROM and local PROM-GPR: same cheap selector reused to keep switching inexpensive and robust.

State Transfer After Cluster Switch (Linear)

Transferred state:

$$\mathbf{u}^s = \mathbf{u}_{0,k^-} + \mathbf{V}_{k^-}\mathbf{q}_{k^-}.$$

If new cluster k^+ uses an affine manifold,

$$\mathbf{u} \approx \mathbf{u}_{0,k^+} + \mathbf{V}_{k^+}\mathbf{q},$$

compute new coordinates by least squares:

$$\mathbf{q}_{k^+} = \arg\min_{\mathbf{q}} \left\| \mathbf{u}^s - (\mathbf{u}_{0,k^+} + \mathbf{V}_{k^+}\mathbf{q}) \right\|_2^2.$$

If $\mathbf{V}_{k^+}^T\mathbf{V}_{k^+} = \mathbf{I}$:

$$\mathbf{q}_{k^+} = \mathbf{V}_{k^+}^T \left(\mathbf{u}_{0,k^-} + \mathbf{V}_{k^-}\mathbf{q}_{k^-} - \mathbf{u}_{0,k^+} \right).$$

State Transfer After Cluster Switch (Nonlinear)

If new cluster k^+ uses a nonlinear manifold

$$\mathbf{u} \approx \Phi_{k^+}(\mathbf{q}),$$

the switch projection is nonlinear least squares:

$$\mathbf{q}_{k^+} = \arg\min_{\mathbf{q}} \|\mathbf{u}^s - \Phi_{k^+}(\mathbf{q})\|_2^2 .$$

With

$$\mathbf{J}_\Phi(\mathbf{q}) := \frac{\partial \Phi_{k^+}}{\partial \mathbf{q}},$$

a Gauss–Newton step at iterate $\mathbf{q}^{(j)}$ is

$$\left(\mathbf{J}_\Phi^T \mathbf{J}_\Phi\right) \Delta\mathbf{q} = \mathbf{J}_\Phi^T \left(\mathbf{u}^s - \Phi_{k^+}(\mathbf{q}^{(j)})\right), \qquad \mathbf{q}^{(j+1)} = \mathbf{q}^{(j)} + \Delta\mathbf{q}.$$

At each switch:

- linear local model $\Rightarrow$ linear least-squares transfer,
- nonlinear local model $\Rightarrow$ nonlinear least-squares transfer.

Online Performance (250×250): $\boldsymbol{\mu} = (4.56, 0.019)$

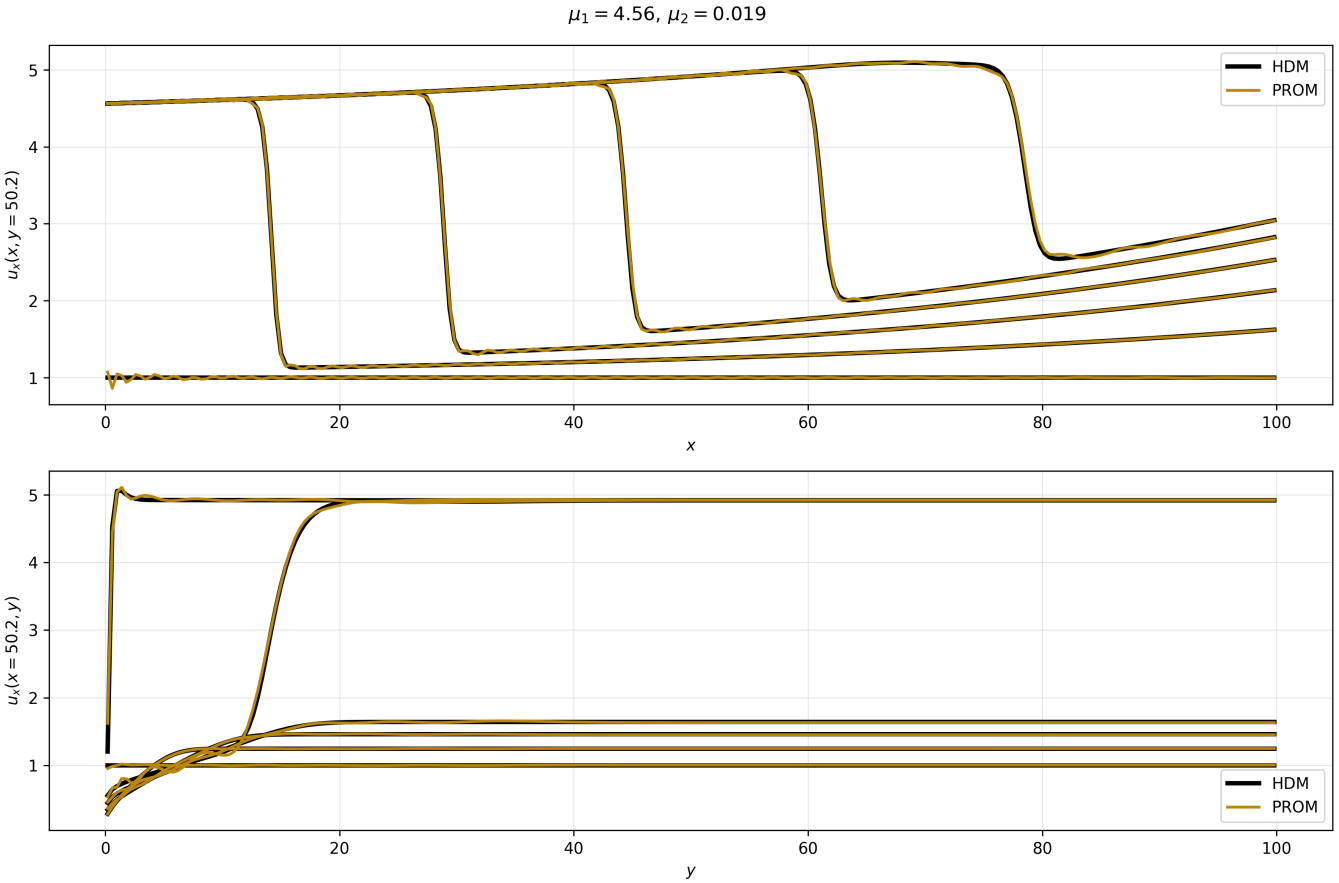
Computational model	n (N for HDM)	$\bar{n}$ (or n_{tot} for POD-DL)	$\mathbb{RE}_{2,\mathbf{u}}$ (%)	Wall-clock time (min)	Speedup
HDM	125 000	–	–	12.512	–
PROM	151	–	0.450	5.934	2.108
QPROM	19	–	4.707	4.731	2.645
PROM-GPR	10	141	2.112	2.374	5.269
PROM-POD-DL	5	$n_{\mathrm{tot}} = 151$	1.242	2.116	5.914
Local PROM ($N_c = 10$)	24–47	–	0.195	3.929	3.184
Local QPROM ($N_c = 10$)	6–9	–	2.605	1.806	6.928
Local PROM-GPR ($N_c = 10$)	5	24–47	0.823	1.701	7.357

Speedup is computed with respect to the HDM runtime, $t_{\mathrm{HDM}} = 750.702$ s.

For PROM-POD-DL, $n = 5$ is the latent dimension and $n_{\mathrm{tot}} = 151$ is the decoded coordinate size.

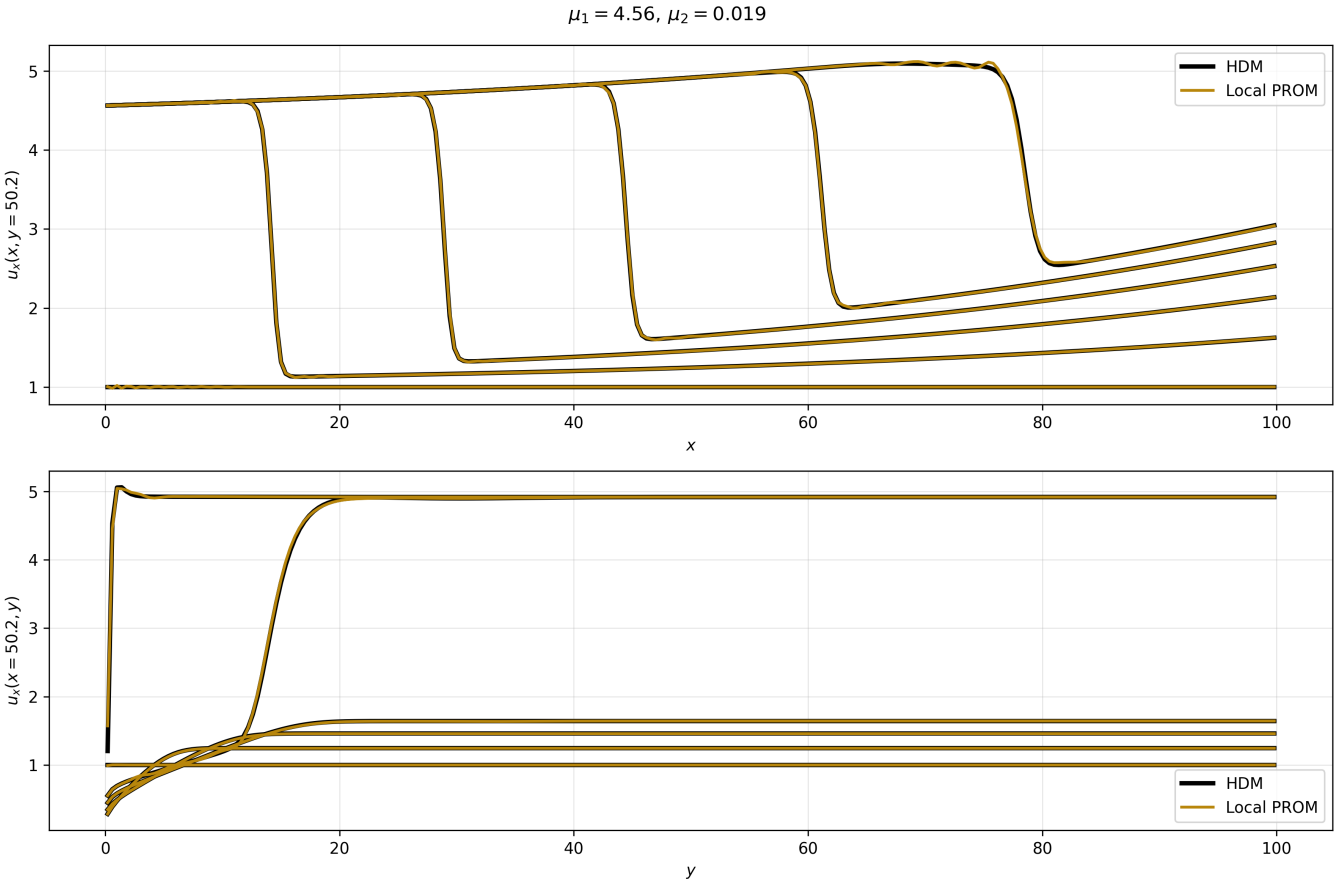
All local models use $N_c = 10$ clusters.

PROM Solution (250×250): $\boldsymbol{\mu} = (4.56,\ 0.019)$



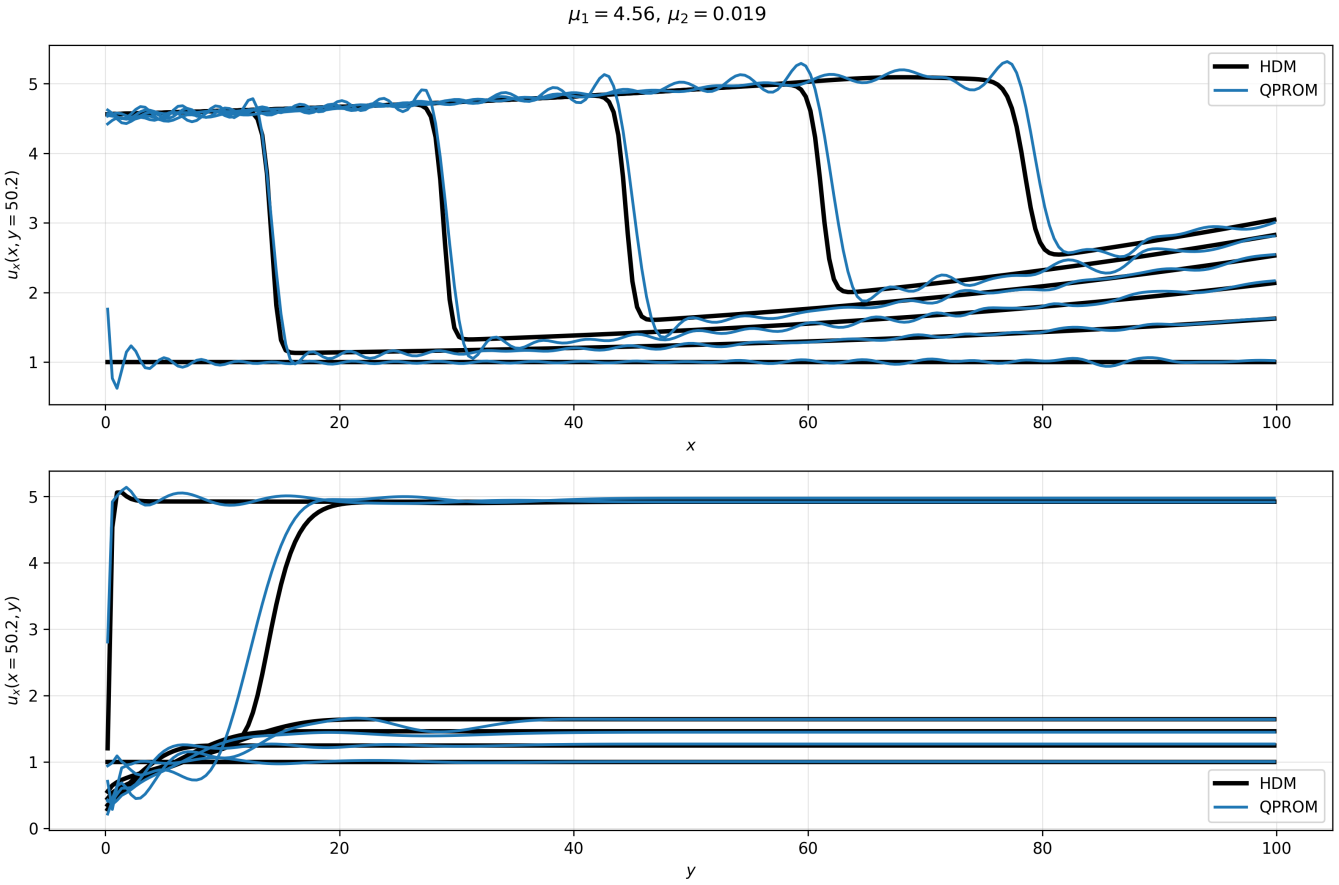
$\mathbb{RE}_{2,\mathbf{u}} = 0.450\%$ $\quad t = 356.059$ s (5.934 min) $\quad$ speedup $= 2.108\times$

Local PROM Solution (250×250): $\boldsymbol{\mu} = (4.56,\ 0.019)$



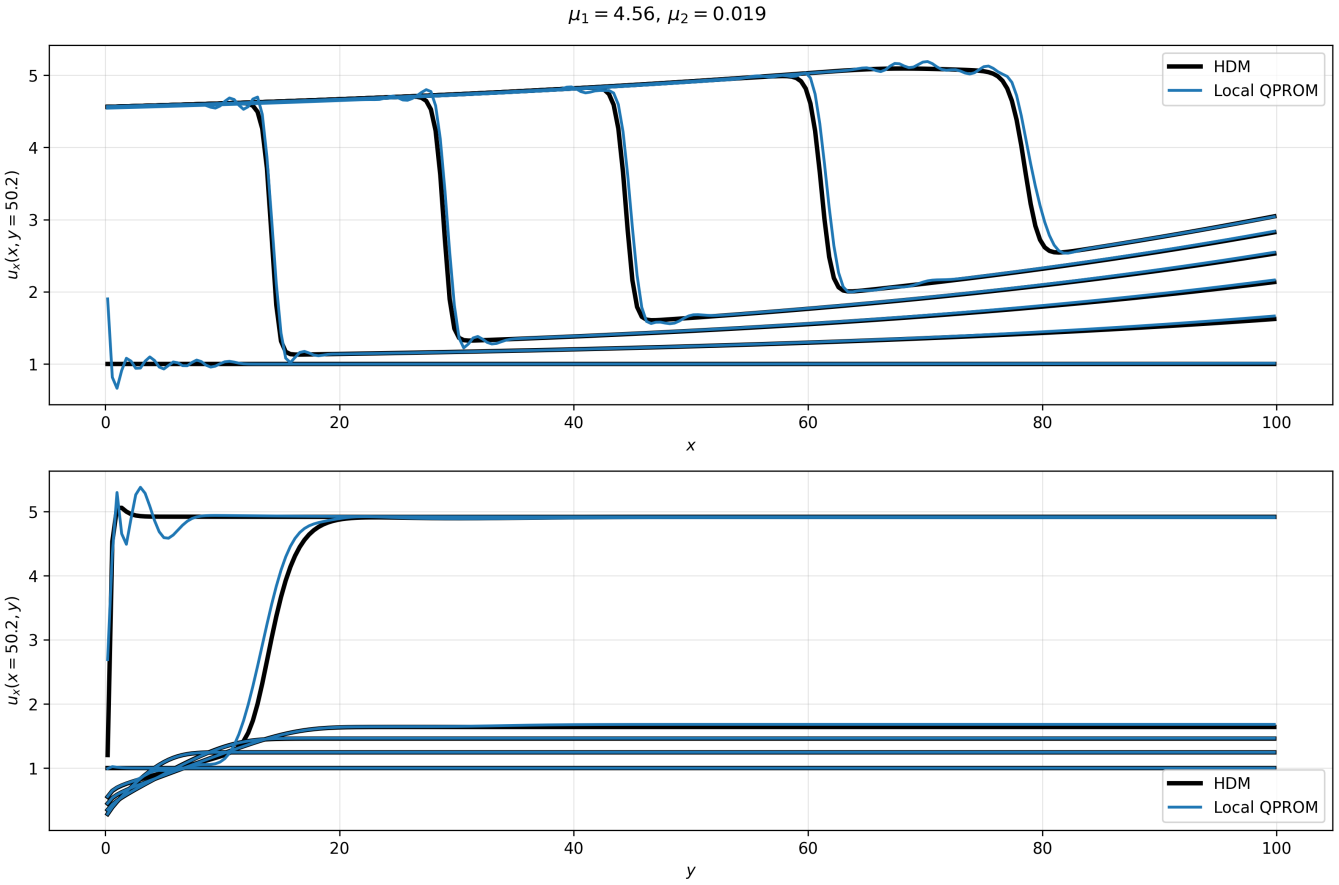
$\mathbb{RE}_{2,\mathbf{u}} = 0.195\%$ $\quad t = 235.750$ s (3.929 min) $\quad$ speedup $= 3.184\times$

QPROM Solution (250×250): $\boldsymbol{\mu} = (4.56,\ 0.019)$



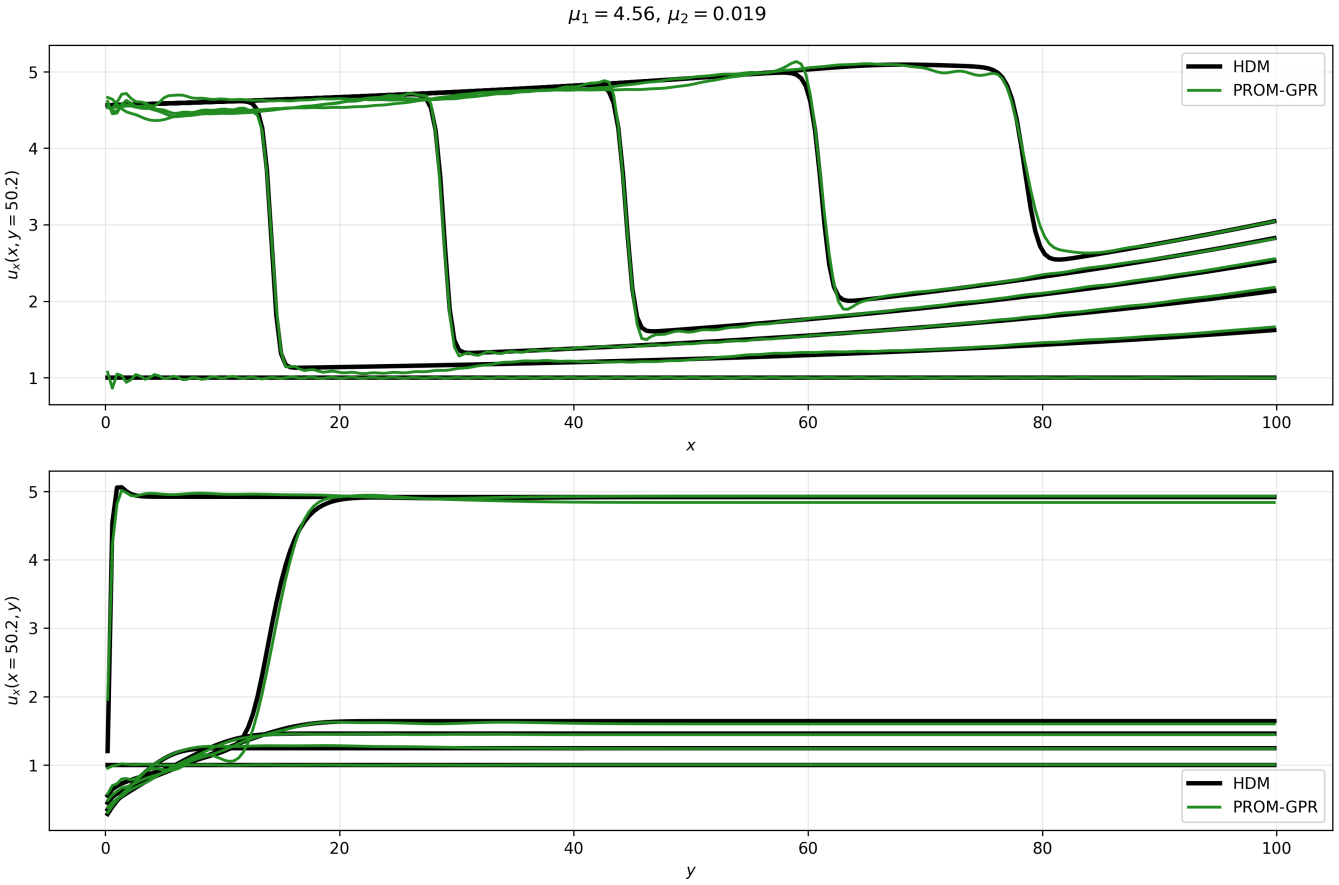
$\mathbb{RE}_{2,\mathbf{u}} = 4.707\%$ $\quad t = 283.871$ s (4.731 min) $\quad$ speedup $= 2.645\times$

Local QPROM Solution (250×250): $\boldsymbol{\mu} = (4.56,\ 0.019)$



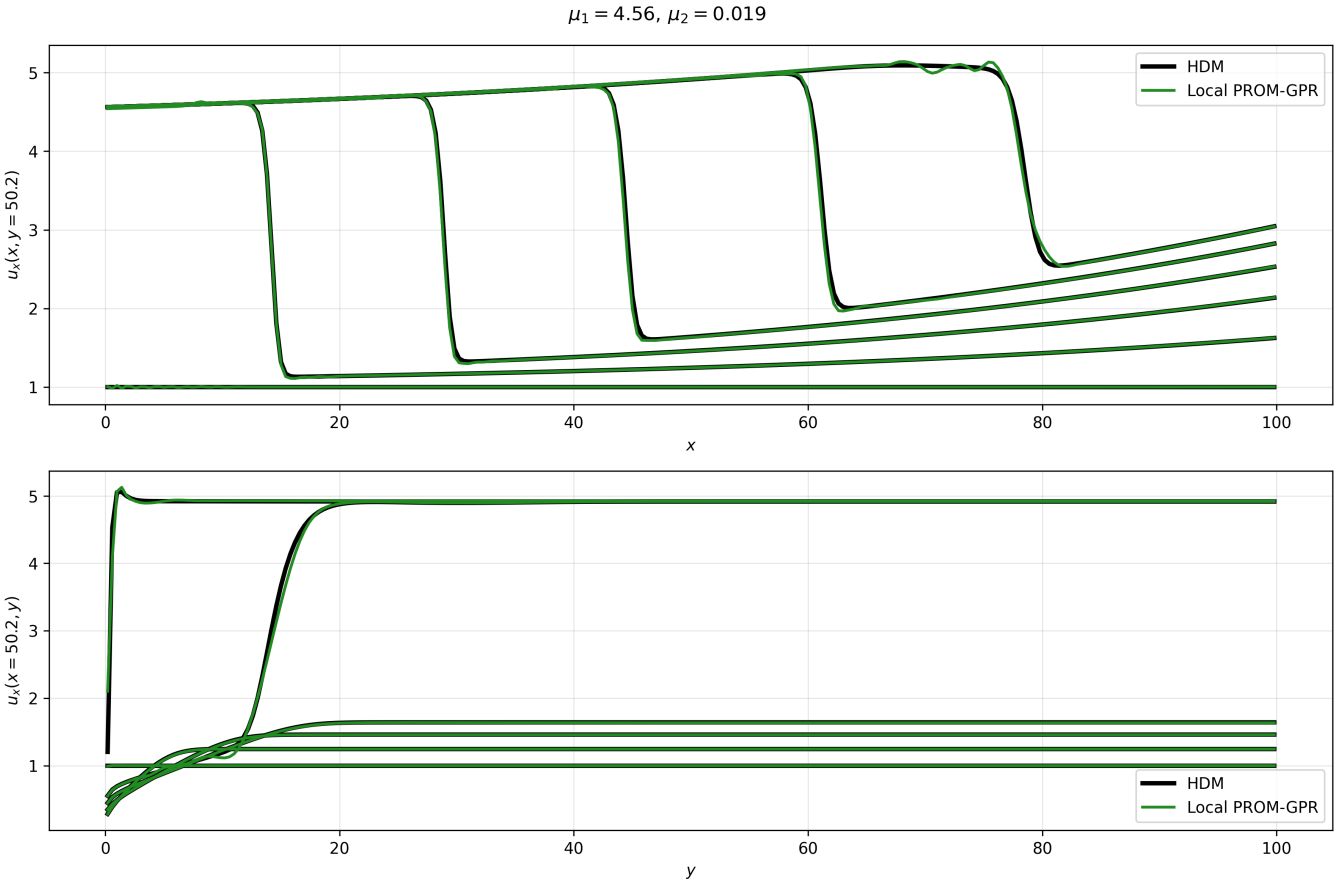
$\mathbb{RE}_{2,\mathbf{u}} = 2.605\%$ $\quad t = 108.365$ s (1.806 min) $\quad$ speedup $= 6.928\times$

PROM-GPR Solution (250×250): $\boldsymbol{\mu} = (4.56,\ 0.019)$



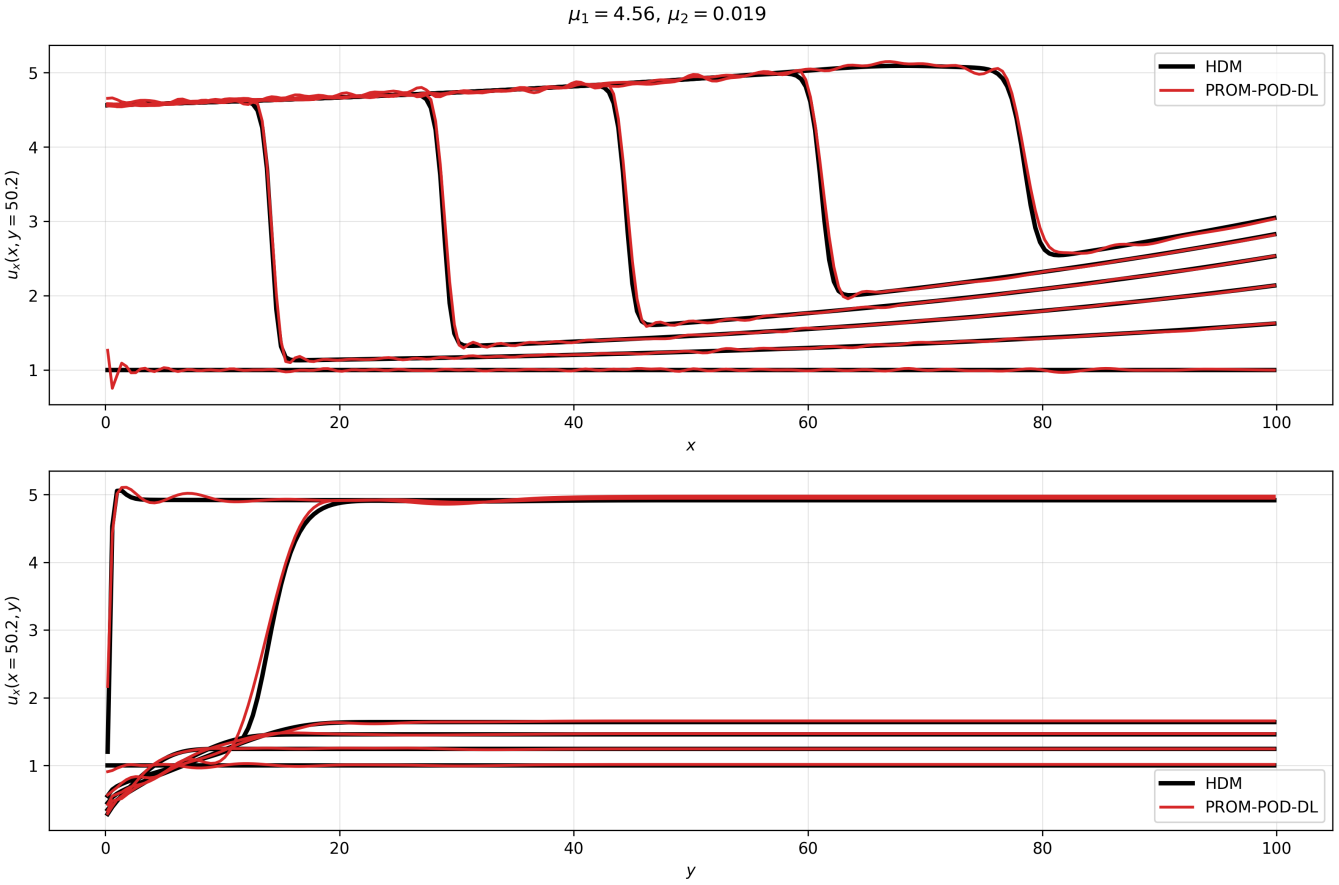
$\mathbb{RE}_{2,\mathbf{u}} = 2.112\%$ $\quad t = 142.463$ s (2.374 min) $\quad$ speedup $= 5.269\times$

Local PROM-GPR Solution (250×250): $\boldsymbol{\mu} = (4.56,\ 0.019)$



$\mathbb{RE}_{2,\mathbf{u}} = 0.823\%$ $\quad t = 102.035$ s (1.701 min) $\quad$ speedup $= 7.357\times$

PROM-POD-DL Solution (250×250): $\boldsymbol{\mu} = (4.56,\ 0.019)$



$\mathbb{RE}_{2,\mathbf{u}} = 1.242\%$ $\quad t = 126.934$ s (2.116 min) $\quad$ speedup $= 5.914\times$

Appendix scope

Main online results use the affine cheap selector from the local linear PROM slides. This appendix records exact extensions for local QPROM and local PROM-GPR. They are included for completeness and are not used in the reported timings/errors.

Local QPROM criterion

Assume current active cluster k^- and reconstructed state

$$\mathbf{u}^s = \mathbf{u}_{0,k^-} + \mathbf{V}_{k^-}\mathbf{q}_{k^-} + \mathbf{H}_{k^-}(\mathbf{q}_{k^-} \otimes \mathbf{q}_{k^-}).$$

Choose the next cluster by nearest centroid $\mathbf{u}_{c,\ell}$, $\ell = 1, \ldots, N_c$:

$$k^+ = \arg\min_{\ell} \|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2.$$

Define

$$\mathbf{a}_\ell := \mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}, \quad \mathbf{b} := \mathbf{V}_{k^-}\mathbf{q}_{k^-}, \quad \mathbf{h} := \mathbf{H}_{k^-}(\mathbf{q}_{k^-} \otimes \mathbf{q}_{k^-}).$$

Then

$$k^+ = \arg\min_{\ell} \|\mathbf{a}_\ell + \mathbf{b} + \mathbf{h}\|_2^2.$$

Local QPROM selector expansion

$$\|\mathbf{a}_\ell + \mathbf{b} + \mathbf{h}\|_2^2 = \|\mathbf{a}_\ell\|_2^2 + 2\,\mathbf{a}_\ell^T\mathbf{b} + 2\,\mathbf{a}_\ell^T\mathbf{h} + \|\mathbf{b}\|_2^2 + 2\,\mathbf{b}^T\mathbf{h} + \|\mathbf{h}\|_2^2.$$

Terms independent of ℓ can be dropped in the minimization, so

$$k^+ = \arg\min_\ell \left[\|\mathbf{a}_\ell\|_2^2 + 2\,\mathbf{a}_\ell^T\mathbf{b} + 2\,\mathbf{a}_\ell^T\mathbf{h}\right].$$

Equivalent original form:

$$k^+ = \arg\min_\ell \Big[\|\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}\|_2^2 + 2(\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell})^T\mathbf{V}_{k^-}\mathbf{q}_{k^-}$$

$$+2(\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell})^T\mathbf{H}_{k^-}(\mathbf{q}_{k^-} \otimes \mathbf{q}_{k^-})\Big].$$

Local QPROM selector (offline/online form)

Precompute offline

$$\mathbf{g}_{k^-,\ell} := \mathbf{V}_{k^-}^T(\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}), \qquad d_{k^-,\ell} := \|\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}\|_2^2,$$

and define $\mathbf{m}_{k^-,\ell} \in \mathbb{R}^{n^2}$ by

$$\mathbf{m}_{k^-,\ell}^T(\mathbf{q} \otimes \mathbf{q}) := (\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell})^T \mathbf{H}_{k^-}(\mathbf{q} \otimes \mathbf{q}).$$

Then online:

$$k^+ = \arg\min_{\ell} \left(2\,\mathbf{g}_{k^-,\ell}^T \mathbf{q}_{k^-} + 2\,\mathbf{m}_{k^-,\ell}^T(\mathbf{q}_{k^-} \otimes \mathbf{q}_{k^-}) + d_{k^-,\ell} \right).$$

Local QPROM switch after a cluster change

If $k^+ = k^-$, no transfer is needed.
If $k^+ \neq k^-$, the transferred state

$$\mathbf{u}^s = \mathbf{u}_{0,k^-} + \mathbf{V}_{k^-}\mathbf{q}_{k^-} + \mathbf{H}_{k^-}(\mathbf{q}_{k^-} \otimes \mathbf{q}_{k^-})$$

must be re-expressed in the new quadratic local manifold

$$\mathbf{u} \approx \mathbf{u}_{0,k^+} + \mathbf{V}_{k^+}\mathbf{q} + \mathbf{H}_{k^+}(\mathbf{q} \otimes \mathbf{q}).$$

Hence the transfer is

$$\mathbf{q}_{k^+} = \arg\min_{\mathbf{q}} \|\mathbf{u}^s - (\mathbf{u}_{0,k^+} + \mathbf{V}_{k^+}\mathbf{q} + \mathbf{H}_{k^+}(\mathbf{q} \otimes \mathbf{q}))\|_2^2 ,$$

typically solved with Gauss–Newton.

Local PROM-GPR criterion

Assume current active cluster k^- and reconstructed state

$$\mathbf{u}^s = \mathbf{u}_{0,k^-} + \mathbf{V}_{k^-}\mathbf{q}_{k^-} + \bar{\mathbf{V}}_{k^-}\bar{\mathbf{q}}_{k^-}, \qquad \bar{\mathbf{q}}_{k^-} = \mathcal{N}_{\mathrm{GPR},k^-}(\mathbf{q}_{k^-}).$$

Nearest-centroid criterion:

$$k^+ = \arg\min_{\ell=1,\ldots,N_c} \|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2.$$

With $\mathbf{a}_\ell := \mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}$,

$$k^+ = \arg\min_{\ell} \left\|\mathbf{a}_\ell + \mathbf{V}_{k^-}\mathbf{q}_{k^-} + \bar{\mathbf{V}}_{k^-}\bar{\mathbf{q}}_{k^-}\right\|_2^2.$$

Local PROM-GPR selector expansion

$$\|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2 = \|\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}\|_2^2 + 2(\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell})^T \mathbf{V}_{k^-}\mathbf{q}_{k^-}$$
$$+2(\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell})^T \bar{\mathbf{V}}_{k^-}\bar{\mathbf{q}}_{k^-} + \mathbf{q}_{k^-}^T \mathbf{V}_{k^-}^T \mathbf{V}_{k^-}\mathbf{q}_{k^-}$$
$$+2\,\mathbf{q}_{k^-}^T \mathbf{V}_{k^-}^T \bar{\mathbf{V}}_{k^-}\bar{\mathbf{q}}_{k^-} + \bar{\mathbf{q}}_{k^-}^T \bar{\mathbf{V}}_{k^-}^T \bar{\mathbf{V}}_{k^-}\bar{\mathbf{q}}_{k^-}.$$

Dropping terms independent of ℓ:

$$k^+ = \arg\min_{\ell} \Big[\|\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}\|_2^2 + 2(\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell})^T \mathbf{V}_{k^-}\mathbf{q}_{k^-}$$
$$+2(\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell})^T \bar{\mathbf{V}}_{k^-}\bar{\mathbf{q}}_{k^-}\Big].$$

Local PROM-GPR selector (offline/online form)

Precompute offline

$$\mathbf{g}_{k^-,\ell} := \mathbf{V}_{k^-}^T(\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}), \qquad \bar{\mathbf{g}}_{k^-,\ell} := \bar{\mathbf{V}}_{k^-}^T(\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}),$$

$$d_{k^-,\ell} := \|\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}\|_2^2.$$

Then online:

$$k^+ = \arg\min_{\ell} \left(2\,\mathbf{g}_{k^-,\ell}^T \mathbf{q}_{k^-} + 2\,\bar{\mathbf{g}}_{k^-,\ell}^T \bar{\mathbf{q}}_{k^-} + d_{k^-,\ell}\right).$$

with

$$\bar{\mathbf{q}}_{k^-} = \mathcal{N}_{\mathrm{GPR},k^-}(\mathbf{q}_{k^-}).$$

Local PROM-GPR switch after a cluster change

If $k^+ = k^-$, no transfer is needed.
If $k^+ \neq k^-$, the transferred state

$$\mathbf{u}^s = \mathbf{u}_{0,k^-} + \mathbf{V}_{k^-}\mathbf{q}_{k^-} + \bar{\mathbf{V}}_{k^-}\mathcal{N}_{\mathrm{GPR},k^-}(\mathbf{q}_{k^-})$$

must be re-expressed in the new local manifold

$$\mathbf{u} \approx \mathbf{u}_{0,k^+} + \mathbf{V}_{k^+}\mathbf{q} + \bar{\mathbf{V}}_{k^+}\mathcal{N}_{\mathrm{GPR},k^+}(\mathbf{q}).$$

Hence the transfer is

$$\mathbf{q}_{k^+} = \arg\min_{\mathbf{q}} \left\| \mathbf{u}^s - \left(\mathbf{u}_{0,k^+} + \mathbf{V}_{k^+}\mathbf{q} + \bar{\mathbf{V}}_{k^+}\mathcal{N}_{\mathrm{GPR},k^+}(\mathbf{q})\right) \right\|_2^2,$$

typically solved with Gauss–Newton.